Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #6

[1] A two-dimensional sequence is given by $P_{X,Y}[i,j] = c(1-p_1)^i(1-p_2)^j$ $i=1,2,\cdots$; $j=1,2,\cdots$ where $0<p_1<1$, $0<p_2<1$ and c is a constant. Find c to make $P_{X,Y}$ a valid joint PMF.

Solution: $1 = \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} c(1-p_1)^i(1-p_2)^j = c\left[\sum_{i=1}^{\infty}(1-p_1)^i\right]\left[\sum_{j=1}^{\infty}(1-p_2)^j\right] = c\left[\frac{1-p_1}{1-(1-p_1)}\right]\left[\frac{1-p_2}{1-(1-p_2)}\right] \Rightarrow c = \frac{p_1p_2}{(1-p_1)(1-p_2)}$

Hence, $P_{X,Y}[i,j] = p_1(1-p_1)^{i-1}\,p_2(1-p_2)^{j-1}$. ☒

[2]. A joint PMF is given as $P_{X,Y}[i,j] = (1/2)^{i+j}$ for $i=1,2,\cdots$, $j=1,2,\cdots$.
If $A=\{(i,j): 1\le i\le 3;\ j\ge 2\}$, find $P[A]$.

Solution: $P[A] = \sum_{(i,j)\in A}\sum P_{X,Y}[i,j] = \sum_{i=1}^{3}\sum_{j=2}^{\infty}\left(\frac{1}{2}\right)^{i+j} = \left[\sum_{i=1}^{3}\left(\frac{1}{2}\right)^i\right]\left[\sum_{j=2}^{\infty}\left(\frac{1}{2}\right)^j\right]$

$= \left[\frac{1}{2}+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3\right]\cdot\left[\frac{1/4}{1-\frac{1}{2}}\right] = \frac{7}{8}\cdot\frac{1}{2} = \frac{7}{16}$ ☒

[3]. The values of a joint PMF are given in the table below. Determine the marginal PMFs.

	$j=0$	$j=1$	$j=2$
$i=0$	1/8	0	1/4
$i=1$	0	1/8	1/4
$i=2$	1/8	0	1/8

Solution: $P_X[i=0] = P_{X,Y}[i=0,j=0]+P_{X,Y}[i=0,j=1]+P_{X,Y}[i=0,j=2] = \frac{1}{8}+0+\frac{1}{4} = \frac{3}{8}$

Similarly, $P_X[i=1] = P_{X,Y}[1,0]+P_{X,Y}[1,1]+P_{X,Y}[1,2] = 0+\frac{1}{8}+\frac{1}{4} = \frac{3}{8}$

$P_X[i=2] = P_{X,Y}[2,0]+P_{X,Y}[2,1]+P_{X,Y}[2,2] = \frac{1}{8}+0+\frac{1}{8} = \frac{2}{8}$

$P_Y[0] = P_{X,Y}[0,0]+P_{X,Y}[1,0]+P_{X,Y}[2,0] = \frac{1}{8}+0+\frac{1}{8} = \frac{2}{8}$

$P_Y[1] = P_{X,Y}[0,1]+P_{X,Y}[1,1]+P_{X,Y}[2,1] = 0+\frac{1}{8}+0 = \frac{1}{8}$

$P_Y[2] = P_{X,Y}[0,2]+P_{X,Y}[1,2]+P_{X,Y}[2,2] = \frac{1}{4}+\frac{1}{4}+\frac{1}{8} = \frac{5}{8}$ ☒

[4]. If a joint PMF is given by $P_{X,Y}[i,j] = p^2(1-p)^{i+j-2}$ $i=1,2,\cdots$; $j=1,2,\cdots$, find the marginal PMFs.

Solution: $P_X[i] = \sum_{j=1}^{\infty}P_{X,Y}[i,j] = \sum_{j=1}^{\infty}p^2(1-p)^{i+j-2} = p^2(1-p)^{i-2}\sum_{j=1}^{\infty}(1-p)^j = p^2(1-p)^{i-2}\,\frac{1-p}{p} = p(1-p)^{i-1}$, $i=1,2,\cdots$

$P_Y[j] = \sum_{i=1}^{\infty}P_{X,Y}[i,j] = \sum_{i=1}^{\infty}p^2(1-p)^{i+j-2} = p^2(1-p)^{j-2}\sum_{i=1}^{\infty}(1-p)^i = p^2(1-p)^{j-2}\cdot\frac{1-p}{p} = p(1-p)^{j-1}$, $j=1,2,\cdots$

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[5] Prove that if the random variables X and Y are independent, then the joint CDF factors as $F_{X,Y}(x,y) = F_X(x)\,F_Y(y)$.

Solution: $F_{X,Y}(x,y) = P[X \le x,\ Y \le y] = \sum_{\{(x_i,y_j):\, x_i \le x,\, y_j \le y\}} \sum P_{X,Y}(x_i,y_j) = \sum_{\{x_i : x_i \le x\}} \sum_{\{y_j :\, y_j \le y\}} P_{X,Y}(x_i,y_j)$

$$= \sum_{\{x_i :\, x_i \le x\}} \sum_{\{y_j :\, y_j \le y\}} P_X(x_i)\,P_Y(y_j)$$

$$= \Big[\sum_{\{x_i :\, x_i \le x\}} P_X(x_i)\Big]\Big[\sum_{\{y_j :\, y_j \le y\}} P_Y(y_j)\Big] = F_X(x)\,F_Y(y).$$

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[6] A joint PMF is given by

$$P_{X,Y}[i,j] = \begin{cases} a, & (i,j) = (0,0) \\ b, & (i,j) = (0,1) \\ c, & (i,j) = (1,0) \\ d, & (i,j) = (1,1) \end{cases}$$

where $a+b+c+d = 1$. Show that a necessary condition for the random variables to be independent is $ad = bc$

Solution:

$P_X[0] = P_{X,Y}[0,0] + P_{X,Y}[0,1] = a+b$ $\qquad$ $P_Y[0] = P_{X,Y}[0,0] + P_{X,Y}[1,0] = a+c$

$P_X[1] = P_{X,Y}[1,0] + P_{X,Y}[1,1] = c+d = 1-(a+b)$ $\qquad$ $P_Y[1] = P_{X,Y}[0,1] + P_{X,Y}[1,1] = b+d = 1-(a+c)$

For independence $P_{X,Y}[i,j] = P_X[i]\,P_Y[j]$

$$\left.\begin{aligned} a &= (a+b)\cdot(a+c) \\ b &= (a+b)\cdot(b+d) \\ c &= (c+d)\cdot(a+c) \\ d &= (c+d)\cdot(b+d) \end{aligned}\right\} \quad \text{use} \quad \begin{aligned} &a+b+c+d = 1 \\ &c+d = 1-(a+b) \\ &b+d = 1-(a+c) \end{aligned} \Rightarrow \frac{a}{b} = \frac{c}{d} \text{ or } ad = bc.$$

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[7] Prove that (for discrete) random variables

(a) $E_{X,Y}[g(X)] = E_X[g(X)]$

(b) $E_{X,Y}[ag(X) + bh(Y)] = a\,E_X[g(X)] + b\,E_Y[h(Y)]$

Solution: (a) $E_{X,Y}[g(X)] = \sum_i \sum_j g(x_i)\,P_{X,Y}(x_i,y_j) = \sum_i g(x_i)\underbrace{\Big\{\sum_j P_{X,Y}(x_i,y_j)\Big\}}_{P_X(x_i)} = \sum_i g(x_i)\,P_X(x_i) = E_X[g(X)]$

(b) $E_{X,Y}[ag(X)+bh(Y)] = \sum_i \sum_j \{a\,g(x_i) + b\,h(y_j)\}\,P_{X,Y}(x_i,y_j)$

$$= \sum_i \sum_j a\,g(x_i)\,P_{X,Y}(x_i,y_j) + \sum_i \sum_j b\,h(y_j)\,P_{X,Y}(x_i,y_j)$$

$$= a\Big[\sum_i g(x_i)\underbrace{\Big\{\sum_j P_{X,Y}(x_i,y_j)\Big\}}_{P_X(x_i)}\Big] + b\Big[\sum_j h(y_j)\underbrace{\Big\{\sum_i P_{X,Y}(x_i,y_j)\Big\}}_{P_Y(y_j)}\Big]$$

$$= a\sum_i g(x_i)\,P_X(x_i) + b\sum_j h(y_j)\,P_Y(y_j)$$

$$= a\,E_X[g(X)] + b\,E_Y[h(Y)]$$

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[8] If X and Y are independent, prove that $E_{X,Y}[g(X)h(Y)] = E_X[g(X)]\cdot E_Y[h(Y)]$

Solution: $E_{X,Y}[g(X)h(Y)] = \sum_i \sum_j g(x_i)h(y_j)\,P_{X,Y}(x_i,y_j) = \sum_i \sum_j g(x_i)h(y_j)\,P_X(x_i)\,P_Y(y_j)$

$$= \Big(\sum_i g(x_i)\,P_X(x_i)\Big)\Big(\sum_j h(y_j)\,P_Y(y_j)\Big)$$

$$= E_X[g(X)]\cdot E_Y[h(Y)]$$

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[9] Find the covariance for the following joint PMF

	$j=0$	$j=1$
$i=0$	1/4	1/4
$i=1$	1/4	1/4

Solution:

$$P_X[0] = \frac{1}{4}+\frac{1}{4} = \frac{1}{2} \qquad P_Y[0] = \frac{1}{4}+\frac{1}{4} = \frac{1}{2}$$
$$P_X[1] = \frac{1}{4}+\frac{1}{4} = \frac{1}{2} \qquad P_Y[1] = \frac{1}{4}+\frac{1}{4} = \frac{1}{2}$$

$$Cov(X,Y) = E_{X,Y}[XY] - E_X[X].E_Y[Y]$$
$$E_X[X] = \sum_{i=0}^{1} i\, P_X[i] = 0\left(\frac{1}{2}\right)+1.\left(\frac{1}{2}\right) = \frac{1}{2}$$
$$E_Y[Y] = \sum_{j=0}^{1} j\, P_Y[j] = 0\left(\frac{1}{2}\right)+1.\left(\frac{1}{2}\right) = \frac{1}{2}$$
$$E_{X,Y}[XY] = \sum_{i=0}^{1}\sum_{j=0}^{1} ij\, P_{X,Y}[i,j] = 0.0\left(\frac{1}{4}\right)+0.1\left(\frac{1}{4}\right)+1.0\left(\frac{1}{4}\right)+1.1\left(\frac{1}{4}\right) = \frac{1}{4}$$
$$Cov(X,Y) = \frac{1}{4} - \frac{1}{2}\times\frac{1}{2} = \frac{1}{4}-\frac{1}{4} = 0.$$

[10] Prove the following properties of the covariance

(a) $cov(X,Y) = E_{X,Y}[XY] - E_X[X].E_Y[Y]$
(b) $Cov(X,X) = var(X)$
(c) $cov(Y,X) = cov(X,Y)$
(d) $Cov(cX,Y) = c[cov(X,Y)] = cov[X,cY]$
(e) $Cov(X,X+Y) = Cov(X,X) + cov(X,Y)$

Solution: (a) $cov(X,Y) = E_{X,Y}[(X-E_X(x))(Y-E_Y[Y])]$

$$= E_{X,Y}[XY - XE_Y[Y] - E_X[X]Y + E_X[X]E_Y[Y]]$$
$$= E_{X,Y}[XY] - \underbrace{E_{X,Y}[X]}_{E_X(X)}E_Y[Y] - E_X[X]\underbrace{E_{X,Y}[Y]}_{E_Y[Y]} + E_X[X]E_Y[Y]$$
$$= E_{X,Y}[XY] - E_X[X]E_Y[Y] - E_X[X]E_Y[Y] + E_X[X]E_Y[Y]$$
$$= E_{X,Y}[XY] - E_X[X]E_Y[Y]$$

(b) $Cov(X,X) = E_{X,X}[(X-E_X(x))(X-E_Y(x))] = E_X[(X-E_X(x))^2] = var[X]$

(c) $Cov(Y,X) = E_{X,Y}[(Y-E_Y[Y]).(X-E_X[X])] = E_{X,Y}[(X-E_X[X]).(Y-E_Y[Y])] = cov(X,Y)$

(d) $Cov(cX,Y) = E_{X,Y}[(cX-E_X[cX])(Y-E_Y[Y])]$
$$= E_{X,Y}[c(X-E_X[X])(Y-E_Y[Y])] = c\,E_{X,Y}[(X-E_X[X])(Y-E_Y[Y])] = c\,cov(X,Y)$$

Similarly $cov(X,cY) = c\,cov(X,Y)$

(e) $Cov(X,X+Y) = E_{X,Y}[(X-E_X(x))(X+Y-E_{X,Y}[X+Y])]$
$$= E_{X,Y}[(X-E_X(x))(X-\underbrace{E_{X,Y}[X]}_{E_X[X]}+Y-\underbrace{E_{X,Y}[Y]}_{E_Y[Y]})]$$
$$= E_{X,Y}[(X-E_X[X])(X-E_X[X])] + E_{X,Y}[(X-E_X[X])(Y-E_Y[Y])]$$
$$= cov(X,X) + cov(X,Y)$$

[11] If X and Y have a covariance of $\text{cov}(X,Y)$, we can transform them to a new pair of random variables whose covariance is zero. To do so we let

$$W = X, \qquad Z = aX + Y, \qquad a = -\frac{\text{cov}(X,Y)}{\text{var}(X)}$$

Show that $\text{cov}(W,Z) = 0$. This process is called <u>decorrelating the random variables</u>.

<u>Solution:</u>
$$\begin{aligned}\text{cov}(W,Z) &= \text{cov}(X, aX+Y) = \text{cov}(X, aX) + \text{cov}(X,Y) \\ &= a\,\text{cov}(X,X) + \text{cov}(X,Y) \\ &= -\frac{\text{cov}(X,Y)}{\text{var}(X)} \cdot \text{var}(X) + \text{cov}(X,Y) = -\text{cov}(X,Y) + \text{cov}(X,Y) = 0.\end{aligned}$$

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[12] If the joint PMF is given as $P_{X,Y}[i,j] = \left(\frac{1}{2}\right)^{i+j}$ $i = 1,2,\ldots$; $j = 1,2,\ldots$, compute the covariance.

<u>Solution:</u>
$$P_{X,Y}[i,j] = \left(\frac{1}{2}\right)^{i+j} = \underbrace{\left(\frac{1}{2}\right)^{i}}_{P_X[i]} \cdot \underbrace{\left(\frac{1}{2}\right)^{j}}_{P_Y[j]} = P_X[i]\,P_Y[j]$$

Hence, X and Y are independent $\Rightarrow \text{cov}(X,Y) = 0$. ☒

[13] Prove that the transformed random variable $\frac{X - E_X[X]}{\sqrt{\text{var}(X)}}$ has an expected value of 0 and variance 1.

<u>Solution:</u>
$$E_X\left[\frac{X - E_X[X]}{\sqrt{\text{var}(X)}}\right] = \frac{1}{\sqrt{\text{var}(X)}} E_X\left[X - E_X[X]\right] = \frac{1}{\sqrt{\text{var}(X)}}\left[E_X[X] - E_X[X]\right] = 0.$$

$$\text{var}\left(\frac{X - E_X[X]}{\sqrt{\text{var}(X)}}\right) = \text{var}\left(\frac{X}{\sqrt{\text{var}(X)}} - \underbrace{\frac{E_X[X]}{\sqrt{\text{var}(X)}}}_{\text{constant}}\right) = \text{var}\left(\frac{X}{\sqrt{\text{var}(X)}}\right) = \frac{1}{\text{var}(X)}\text{var}(X) = 1.$$

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[14] Prove that if $Y = aX + b$, where a and b are constants, then $\rho_{X,Y} = 1$ if $a > 0$ and $\rho_{X,Y} = -1$ if $a < 0$.

<u>Solution:</u>
$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} = \frac{\text{cov}(X, aX+b)}{\sqrt{\text{var}(X)\cdot\text{var}(aX+b)}} = \frac{\text{cov}(X,aX) + \text{cov}(X,b)}{\sqrt{\text{var}(X)\cdot a^2\text{var}(X)}}$$

$$= \frac{a\,\text{cov}(X,X) + \text{cov}(X,b)}{|a|\cdot\text{var}(X)} = \frac{a\cdot\text{var}(X) + \text{cov}(X,b)}{|a|\cdot\text{var}(X)}$$

$$\text{cov}(X,b) = E_X\left[(X - E_X[X])\cdot(b - E[b])\right] = E_X\left[(X - E_X[X])\cdot 0\right] = 0 \cdot E_X\left[(X - E_X[b])\right] = 0.$$

Hence, $\rho_{X,Y} = \frac{a\cdot\text{var}(X)}{|a|\cdot\text{var}(X)} = \frac{a}{|a|} = \begin{cases} 1, & a > 0 \\ -1, & a < 0. \end{cases}$

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[15] In this problem, we compare the prediction of a random variable with and without the knowledge of a second random variable outcome. Conside the joint PMF shown below. First determine the optimal linear prediction of Y without any knowledge of the outcome of X. Also, compute the minimum mean square error. Next, determine the optimal

linear prediction of Y based on the knowledge that $X=x$ and compute minimum mean square error. Plot the predictions versus x in the plane. How do the minimum mean square errors compare?

	$j=0$	$j=1$
$i=0$	$\frac{1}{8}$	$\frac{1}{4}$
$i=1$	$\frac{1}{4}$	$\frac{3}{8}$

Solution:

<u>without knowledge of X</u>

optimal prediction $\hat{Y} = E_Y[Y]$

Minimum mean square error $mse_{min} = var(Y)$

$P_Y[0] = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$ $P_Y[1] = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}$ $\Rightarrow$ $E_Y[Y] = 0 \cdot \frac{3}{8} + 1 \cdot \frac{5}{8} = \frac{5}{8}$

$var[Y] = [0^2 \cdot \frac{3}{8} + 1^2 \cdot \frac{5}{8}] - (\frac{5}{8})^2 = \frac{5}{8} - \frac{25}{64} = \frac{15}{64}$

Hence, $\boxed{\hat{Y} = \frac{5}{8}}$, $\boxed{mse_{min} = \frac{15}{64}}$

<u>with knowledge of X</u>

optimal prediction $\hat{Y} = E_Y[Y] + \frac{cov(X,Y)}{var(X)}[X - E_X[X]]$

$P_X[0] = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$ $P_X[1] = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}$ $\Rightarrow$ $E_X[X] = \frac{5}{8}$

$var(X) = var(Y) = \frac{15}{64}$

$E_{X,Y}[XY] = 0.0(\frac{1}{8}) + 0.1(\frac{1}{4}) + 1.0 \cdot (\frac{1}{4}) + 1.1 \cdot (\frac{3}{8}) = \frac{3}{8}$

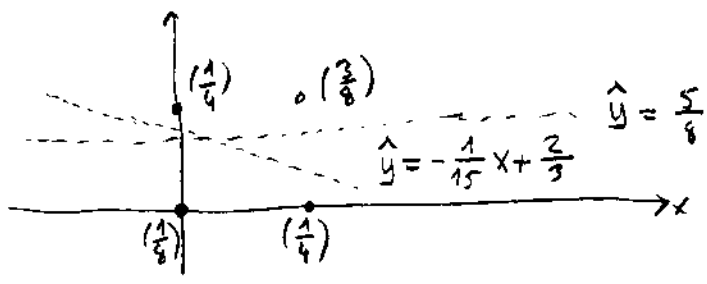
$Cov(X,Y) = E_{X,Y}[XY] - E_X[X] \cdot E_Y[Y] = \frac{3}{8} - \frac{5}{8} \cdot \frac{5}{8} = -\frac{1}{64}$

Hence, $\hat{Y} = \frac{5}{8} - \frac{1/64}{15/64}[X - \frac{5}{8}] = \boxed{-\frac{1}{15}X + \frac{2}{3}}$

Minimum mean square error $mse_{min} = var(Y)(1 - \rho_{X,Y}^2)$

$\rho_{X,y} = \frac{cov(X,Y)}{\sqrt{var(X) \cdot var(Y)}} = \frac{-1/64}{\sqrt{\frac{15}{64} \cdot \frac{15}{64}}} = -\frac{1}{15}$

$mse_{min} = \frac{15}{64}(1 - (-\frac{1}{15})^2) = \boxed{\frac{7}{30} < \frac{15}{64}}$



[16] Consider $\text{var}(X+Y)$, where X and Y are correlated random variables ($\rho_{X,Y} \neq 0$). How is the variance of a sum of random variables affected by the correlation between the random variables? (Hint: express the sum of the random variables using the correlation coefficient!)

Solution:

$$\begin{aligned}
\text{var}(X+Y) &= \text{var}(X) + \text{var}(Y) + 2\,\text{cov}(X,Y) \\
&= \text{var}(X) + \text{var}(Y) + 2\,\rho_{X,Y}\sqrt{\text{var}(X)\,\text{var}(Y)} \\
&< \text{var}(X) + \text{var}(Y), \quad \text{if } \rho_{X,Y} < 0 \\
&> \text{var}(X) + \text{var}(Y), \quad \text{if } \rho_{X,Y} > 0.
\end{aligned}$$

